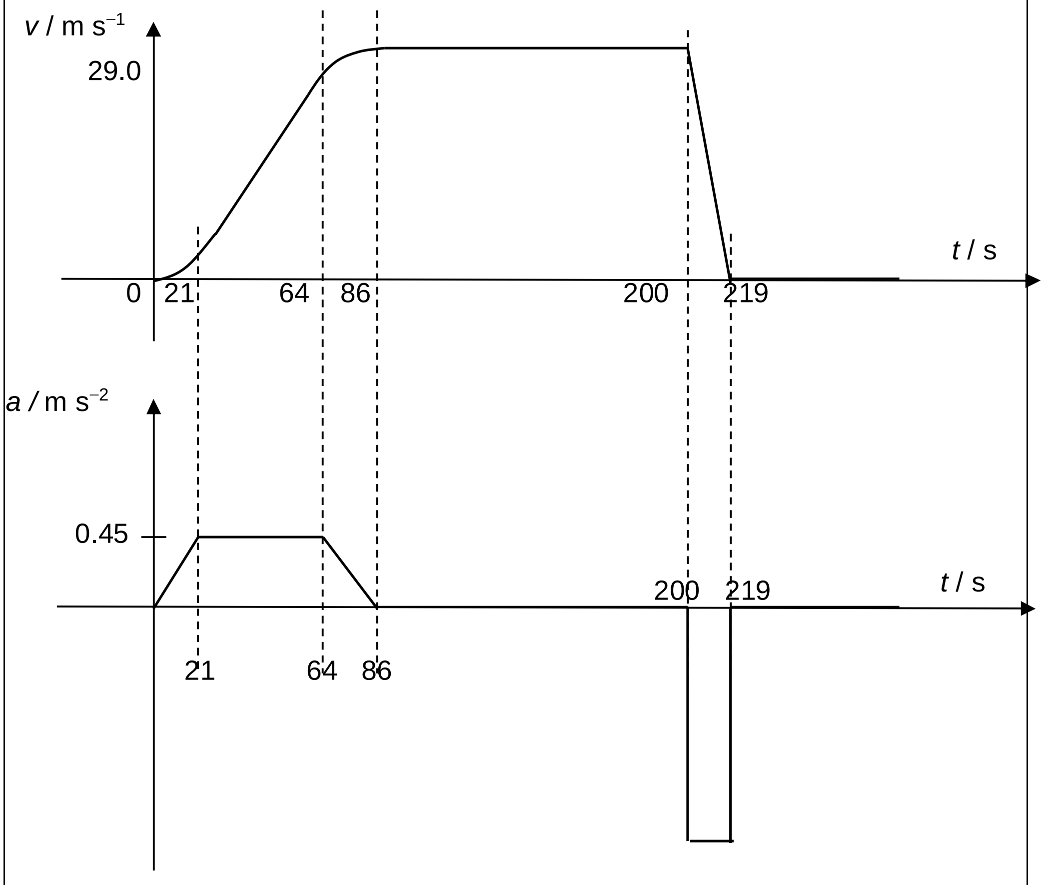
Suggested Solution

Section A

1	**(a)**		Δv = area under *a-t* graph $= \frac{1}{2}(21)(0.45) + (43)(0.45) + \frac{1}{2}(22)(0.45)$ $= 29$ $\langle a \rangle = \Delta v / \Delta t$ $= 29 / 200$ $= 0.15\ (0.145)\ \text{m s}^{-2}$	**1** **1**
	(b)	**(i)**	v_{max} = area under *a-t* graph (from t = 0 to t = 86 s) $= \frac{1}{2}(86 + 43)(0.45)$ $= 29.0\ \text{m s}^{-1}$	**1** **1**
		(ii)		
2	**(a)**		[1] for the correct pair of forces	